

Optimized mix design of rice husk ash blended high-strength concrete through machine learning models

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Abstract

Reducing cement consumption without compromising the mechanical performance of high-strength concrete requires reliable, data-driven mix design approaches. This study presents an integrated experimental and machine-learning-based framework for optimizing rice husk ash (RHA) blended high-strength concrete. A comprehensive experimental dataset was developed covering both fresh properties (slump and compaction factor) and hardened properties (compressive, split tensile, and flexural strengths) across wide ranges of cement content (300–450 kg/m³), water–binder ratio (0.40–0.55), RHA replacement level (0–20 %), and curing age (7–90 days). Multiple machine learning models have been used in this study to simultaneously predict compressive, tensile, and flexural strengths, moving beyond single-property prediction toward a unified performance evaluation. Ensemble-based models and optimized artificial neural networks showed the best predictive performance, achieving coefficients of determination above 0.95 with consistently low prediction errors. Model interpretability was enhanced through feature-importance analysis, which identified cement content and water–binder ratio as the dominant parameters governing strength development, followed by RHA content and curing age. A data-driven multi-objective screening framework was implemented to identify mix proportions that balance mechanical performance, workability, and sustainability. Screening of experimentally tested mixes indicated an optimal RHA replacement range of 10–15 %, providing strength improvements of approximately 10–12 % at later ages while reducing cement consumption by approximately 10–14 %, corresponding to an estimated embodied CO₂ saving of about 35–55 kg CO₂ per m³ of concrete.

Introduction

Concrete production has been recognized as a major source of carbon dioxide emissions because ordinary Portland cement requires high energy and releases large amounts of

greenhouse gases. To lower clinker content while maintaining performance, rice husk ash has been adopted as a supplementary cementitious material because it is abundant, rich in reactive silica, and can be made highly pozzolanic when amorphous and finely ground [1]. When properly processed ash has been used, pore structure has been refined, the interfacial transition zone has been densified, and improvements in mechanical strength and durability have been reported in many test programs [2]. In mortar and concrete systems, increases in later-age compressive strength and reductions in transport properties have been attributed to the reaction of amorphous silica from the ash with calcium hydroxide, which has been shown through microstructural and performance data [3]. Optimal replacement levels have been identified. In self-compacting mixes, replacement in the range of about ten to fifteen percent has been reported to give the best balance of strength and fresh properties, with mixtures outside that range showing either reduced early strength or excessive loss of flow without admixture adjustment [4]. Work that combined nano additives in cementitious systems has shown that permeability can be reduced and mechanical properties can be improved at later ages, which has been associated with additional pore refinement and crack-bridging mechanisms [5]. In high performance concrete, fifteen percent replacement has been reported to improve compressive strength and microstructure at later ages, while shrinkage and permeability have been reduced relative to control mixes [6]. For self-compacting concrete made with treated ash and other additions, target flow and passing ability have been achieved with measured strength at early and later ages comparable to mixes without ash when admixture dosage has been optimized [7]. Under elevated temperature exposure, concretes with partial replacement by ash have retained higher residual strength and have shown improved volumetric stability compared with ordinary mixes, especially up to moderate temperatures where re-curing has also been beneficial [8]. In recycled aggregate concrete, the use of reactive ash has been shown to recover strength and improve durability indices compared with recycled mixes without ash [9]. In related systems, freeze–thaw resistance of recycled concrete has been enhanced when ash has been incorporated at moderate dosages, which has been attributed to the refined paste microstructure and improved matrix quality [10]. Because mixture behavior depends on multiple variables that interact in a nonlinear way, data-driven modeling has been adopted to complement testing. A compiled dataset of one hundred ninety-two mixes has been used to train artificial neural networks and adaptive neuro-fuzzy inference systems, and coefficients of determination near zero point nine two have been reported for compressive strength prediction, which has exceeded linear regression performance near zero point eight two on the same data [11]. Interpretable machine learning has also been applied; feature importance analyses have indicated that water to binder ratio has been the dominant predictor, followed by curing age and ash content, when compressive strength of

ash blended concrete has been modeled [12]. Hybrid approaches have been developed in which boosting and regularization have been combined; in one study on sustainable ash concretes, R squared values around zero point nine eight and root mean square error below two mega pascals have been obtained on several hundred mixes [13]. Ensemble learner optimization across multiple algorithms has likewise delivered high accuracy for ash concrete strength prediction, which has demonstrated that tuned ensembles can be used as reliable surrogates for trial batching in preliminary design [14]. A hybrid artificial neural network that was optimized by a reptile search algorithm has achieved R squared near zero point nine seven on a large dataset with varied replacement levels, cement contents, water to binder ratios, and curing ages [15]. Processing conditions for the ash have been identified as critical. Reviews focused on particle properties have concluded that controlled calcination in the range of about six hundred to seven hundred degrees Celsius and fine grinding below ten micrometers have been needed to obtain high pozzolanic activity and to limit unburnt carbon, which has influenced both strength gain and workability response [16]. Beyond neural and boosting models, symbolic learning by genetic expression programming has been used to form closed-form predictive equations for ash concretes; such models have matched measured strengths with low error and have supported parametric studies on mixture variables [17]. At high temperature, systematic testing has shown that partial replacement with ash can raise residual strength and reduce mass loss compared with ordinary mixes, and that re-curing after thermal exposure has further improved recovery, thereby widening the application domain for ash concretes in thermal environments [18]. When glass powder has been combined with ash, rate-dependent testing has reported measurable changes in slump, absorption, and strength with strain rate, which has highlighted that mixture design with multiple supplementary materials can be tuned to control both fresh and dynamic behaviors [19]. Broader characterization work on biomass ashes has also confirmed that oxide composition, loss on ignition, and fineness must be controlled to ensure consistent performance when such ashes are used as cement replacements, and that these parameters directly influence water demand, reactivity, and durability outcomes in concrete [20].

Despite extensive experimental and analytical studies on rice husk ash as a supplementary cementitious material and the increasing use of machine learning techniques in concrete research, several important gaps remain. Most existing studies primarily focus on compressive strength prediction, while tensile and flexural strengths are often examined separately or not modeled within a unified framework. Fresh-state properties such as slump and compaction factor, which are essential for practical mix design and constructability, are rarely integrated alongside mechanical properties in data-driven models. In addition, many machine learning-based investigations rely on relatively limited

datasets with narrow ranges of mix parameters, restricting the robustness and generalizability of their predictions. Furthermore, while high prediction accuracy is frequently reported, limited attention has been given to model interpretability and optimization-based decision support that can translate predictions into actionable mix design guidance. These limitations highlight the need for an integrated framework that combines a comprehensive experimental dataset, simultaneous multi-property prediction using interpretable machine learning models, and multi-objective optimization to support sustainable and practical design of high-strength concrete incorporating rice husk ash. In this study, high-strength concrete is defined as concrete achieving a characteristic compressive strength exceeding 50 MPa at 28 days, based on 150 mm cube specimens, in accordance with commonly adopted international classifications.

The objective of this study is to develop an integrated experimental and machine learning-based framework for the sustainable design of high-strength concrete incorporating rice husk ash (RHA). A comprehensive experimental program was conducted to generate a multi-property dataset covering fresh-state properties (slump and compaction factor) and hardened-state properties (compressive, split tensile, and flexural strengths) across wide ranges of cement content, water-binder ratio, RHA replacement level, and curing age. This dataset was used to evaluate multiple machine learning models for the simultaneous prediction of key mechanical properties, moving beyond the conventional single-property approach. In addition, model interpretability was incorporated through feature-importance analysis to identify the dominant mix parameters governing strength development and workability. Finally, a multi-objective optimization framework was implemented to identify mix proportions that balance mechanical performance, workability, and sustainability by reducing cement consumption.

The significance of this work lies in its unified integration of experimental testing, interpretable machine learning, and optimization-based decision support into a single practical framework for concrete mix design. Unlike most existing studies that focus on isolated strength parameters or purely predictive accuracy, the present approach enables simultaneous multi-property performance assessment while providing mechanistic insight into the influence of mix parameters. The constrained optimization framework further translates model predictions into actionable mix design guidance that supports both performance enhancement and cement reduction. Consequently, the proposed methodology offers a scalable and data-driven pathway for the wider adoption of agricultural waste materials in high-strength and sustainable concrete construction.

Section snippets

Research methodology

The methodology of this study has been designed to integrate experimental data with advanced data-driven modeling techniques for predicting the performance of rice husk ash (RHA) blended high-strength concrete. The approach consists of dataset preparation, experimental testing, data preprocessing, machine learning model development, evaluation metrics, and optimization strategies. The scope of the present study is focused on experimentally measured fresh and mechanical properties and their

Cement

Ordinary Portland Cement of 53 grade was selected to ensure adequate early and later-age strength development suitable for high-strength concrete applications conforming to IS: 8112-1989 [39] and ASTM C150 Type I specifications [40]. The cement had a specific gravity of 3.12, and its properties were determined according to IS: 4031 [41] and ASTM C188 [42]. The physical characteristics are listed in Table 1.

Fine aggregate

Natural river sand obtained from local sources was used as fine aggregate. The sand

Mix-proportioning

The influence of cement replacement with rice husk ash (RHA) was investigated across a wide range of water-binder ratios and binder contents. The relative proportions of coarse and fine aggregates were determined experimentally in the laboratory using the packing density method, which is recognized as a reliable approach for achieving dense aggregate skeletons and optimizing paste requirement in concrete [51,52]. Packing density is defined as the ratio of the volume of solids to the bulk volume

Workability characteristics

The fresh-state properties of RHA-blended concretes were assessed in terms of slump and compaction factor. A consistent reduction in slump was observed with increasing RHA content, irrespective of cement content and w/b ratio (Fig. 2a). For example, at a cement content of 300 kg/m³ and w/b = 0.40, slump decreased from approximately 50 mm in the control mix (0 % RHA) to nearly 20 mm at 20 % RHA replacement. This loss in workability is attributed to the high surface area and porous texture of RHA

Interpretation of machine learning results

To enhance the transparency and practical relevance of the developed machine learning models, interpretability analyses were conducted to understand the influence of input parameters on predicted mechanical properties. This section focuses on extracting meaningful insights from the trained models rather than reiterating data preprocessing or

model training procedures, which have been described earlier. Feature importance analysis and partial dependence evaluation were employed to identify

Checking for outliers

Details of outlier detection and correlation analysis are provided in the Data Preprocessing section; the resulting diagnostic plots (Fig. 6a–b) are referenced here only for interpretive context.

Machine learning model development

In the present study, a set of supervised machine learning models was implemented and systematically evaluated for predicting the mechanical properties of RHA-blended high-strength concrete. The models actually trained and tested include linear regression (LR), decision tree (DT), random forest (RF), support vector regression (SVR), artificial neural network (ANN), gradient boosting regression (GBR), and extreme gradient boosting (XGBoost). These models were selected to represent a range of

Screening-based mix design framework

For screening purposes, workability parameters were not measured experimentally for each candidate mix but were estimated using the trained predictive models. A screening-based, multi-objective decision-support framework was developed to identify feasible combinations of mix design parameters that balance mechanical performance, fresh-state workability, and sustainability using trained machine learning models as surrogate predictors. Rather than implementing a formal evolutionary or genetic

Limitations and uncertainty

While the present study provides a comprehensive experimental and data-driven framework for optimizing rice husk ash (RHA) blended high-strength concrete, certain limitations should be acknowledged. The RHA used in this investigation was obtained from a single source and produced under controlled burning and grinding conditions. Variations in RHA source, combustion temperature, fineness, and chemical composition may influence pozzolanic reactivity and, consequently, concrete performance.

Conclusion

This study advances existing research on rice husk ash-based concrete by moving beyond conventional experimental or predictive analyses to propose a unified, data-driven mix design framework. The key contributions include the development of a comprehensive multi-property experimental dataset, simultaneous prediction of multiple strength

parameters using machine learning, incorporation of interpretability tools to explain model behaviour, and implementation of a constrained screening-based

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Data availability

The datasets generated during the current study are available from the corresponding author on reasonable request.

Ethics approval

Not applicable.

CRediT authorship contribution statement

Anasuya Mondal: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing, Investigation, Data curation, Software, Formal analysis, Visualization, Writing – original draft.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.